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MILITARY SPECIFICATION

TEST SET, ELECTRICAL POWER A/M24T-8( )

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of load bank, electrical power, generator set, mobile test set designated A/M24T-8( ).

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof For Packaging And Sealing

Military

MIL-M-59	Mat, Landing, Pierced-Plank Type
MIL-P-116	Preservation, Methods Of
MIL-D-1000	Drawings, Engineering And Associated Lists
MIL-C-3774	Crates, Wood, Open, 12000 to 16000 Pounds
MIL-G-4008	Generator Set, Gasoline, Engine Driven, Aircraft Support, General Specification For
MIL-E-4158	Electronic Equipment, Ground, General Requirements For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: AFLC CASO/LODS, Federal Center, Battle Creek MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-W-5086	Wire, Electrical, 600 Volt, Copper, Aircraft
MIL-E-5088	Wiring, Aircraft, Installation Of
MIL-E-5220	Extinguisher, Fire, Aircraft, Type A-20
MIL-M-10304	Meters, Electrical, Indicating Panel Type, Ruggedized, General Specification For
MIL-R-12144	Reflector, Indicating, Clearance

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Specifications And Standards, Order Of Precedence For The selection Of
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electronic Interference Characteristics Requirements For Equipment
MIL-STD-462	Electromagnetic Interference Characteristcs, Measurement Of
MIL-STD-470	Maintainability Program Requirements (For Systems And Equipment)
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-808	Finishes, Protective, And Codes, For Finishing Schemes For Ground And Ground Support Equipment
MIL-STD-810	Environmental Test Methods For Aerospace And Ground Equipment
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-889	Dissimilar Metals
MS3057	Adapter, Electrical, Accessory To Cable
MS3101	Connector, Plug, Electric, Cable
MS3106	Connector, Plug, Electric, Straight
MS3108	Connector, Plug, Electric Angle 90 Degree
MS3420	Bushing, Cable Adapter, Telescoping
MS25018	Receptacle-Jet Starting Power
MS51335	Pintle Assembly, Towing - 18000 Pounds Capacity, Manual Release
MS51336	Lunette Coupler, Drawbar Ring

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AIR FORCE - NAVY AERONAUTICAL

AN2552	Receptacle-External Power 28 Volts DC
AN3114	Receptacle, External Power, 115/200 Volts, 3 Phase

DRAWINGS

Air Force

55C6182	Connectors-400 Volt/416 Volt Power Cable

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Handbook H28 Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington DC 20402.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The test set shall consist of the following major components.

Item Name	See Requirement
Load controls	3.7.1
Cooling fans	3.7.2
Enclosure	3.7.3
Control console	3.7.4
Electrical hardware and accessory equipment	3.7.5

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

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3.4 Materials.

* 3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall be selected in accordance with MIL-STD-454, Requirement 4. Treatment with a fungicidal agent except as specified therein, shall be subject to approval by the procuring activity.

* 3.4.2 Metals. Metals shall be of the corrosion resistant type, or suitably processed to resist corrosion due to fuels, salt spray, or atmospheric conditions as may be encountered in storage or normal service. Corrosion of more than a noticeable extent shall not be acceptable. At no time will the corrosion be allowed to interfere with the operation of load bank.

* 3.4.2.1 Dissimilar metals. Dissimilar metals shall be selected in accordance with MIL-STD-889 and MIL-STD-454, Requirement 16.

* 3.4.3 Nonstandard parts. Nonstandard parts shall be selected per MIL-STD-454, Requirement 22. Percent of nonstandard parts shall be predicted and furnished to the procuring activity for approval.

3.5 Design and construction. All parts, such as shafts, bearings, pistons, gears, valves, armatures, regulators, and controls shall have proper clearances and adjustments and shall so work together that the test set will supply the rated requirements without unnecessary strains, vibrations, and overheating.

* 3.5.1 Reliability. The test set shall have a specified mean time between failures (MTBF) of at least 400 hours, tested in accordance with MIL-STD-781.

3.5.2 Maintainability. The load bank shall meet the requirements of MIL-STD-470 with verification in accordance with MIL-STD-471. The load bank shall have a mean corrective maintenance time of 100 minutes and a maximum corrective maintenance time of 300 minutes. The producer's risk shall be 0.10 and the consumer's risk shall be 0.10. It shall be possible to trouble-shoot and repair the load bank by replacement of a component part without the need to remove any other component or a subassembly, except the panel covers, interconnecting cables and the component brackets.

3.5.3 Alternating current impedance loading. The alternating-current impedance load bank shall be so designed that by the use of fixed steps and variable components, indefinitely variable loading on the resistive and reactive sections between zero and the loads specified in 3.5.7 may be obtained at any power factor between 0.75 lagging and unity. The resistance and inductance values presented to the circuit by the resistive and reactive elements shall not be affected by the current through them except for normal heating. No device shall be used that increases distortion of the applied waveform more than 2 percent in terms of root mean square total harmonic content.

3.5.4 Operation. The load bank shall furnish a 3-phase, wye-connected load. Each phase shall be independently adjustable. The values of loading specified herein shall be obtainable without considering the illuminating and control device loads. The load bank shall provide the specified loads at all voltages, frequencies, and power factors specified herein.

3.5.4.1 Single phase loading. The load bank shall have provisions for convenient reconnection of the three phases so that it will be possible to obtain a single-phase load equal to the maximum 3-phase rating. The single-phase load shall be available by connecting the input power to the neutral

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terminal and to a common connection of the 3-phase terminals. When used in this manner, all controls, switches, meters, and protective devices, except the phase rotation indicators, shall function normally. The means of connection shall be such that it will require one man less than 5 minutes to effect the conversion from single-phase to 3-phase or vice versa. Special tools shall not be required for the conversion.

3.5.5 Resistance loading. The resistance load bank shall consist of two sections. Each section, by means of fixed-step components and variable components, shall have the capability of providing any load from 0 to 1,500 amperes (amp) in 1-amp increments.

3.5.5.1 Split bus. Provisions shall be incorporated for testing the split-bus characteristics of generator sets built in accordance with MIL-G-4008. Electrical connections shall be so incorporated that a simulated aircraft-servicing load may be carried on one bank while the other bank carries the simulated jet-engine starting load. The bank to carry the jet-engine starting load shall be so designated.

3.5.6 Jet starting. The test set shall be designed to supply the signal for normal jet starting to the control terminal of the generator cable. The signal shall be applied by means of a momentary contact switch on the control panel. Circuitry to accomplish this shall be essentially as shown on Figure 1.

3.5.7 Electrical properties. Electrical properties shall be adequate to fulfill the following load requirements and shall be the continuous-duty ratings of the units. It shall be possible to use full direct-current load and any one alternating-load simultaneously.

Direct-current loads (resistance only)

Volts	Amperes	Wiring Method	Function
28	0-1,500	Wired for split bus	Aircraft-servicing load
28	0-1,500	Wired for split bus	Jet-Engine starting load

Alternating-current loads (resistance and inductive reactance)

CPS	Volts	Phase	KW	KVA	KVAR	PF Lag
400	120/208	3, 4-wire	0-120	0-160	0-106	.75-1.0
60	120/208	3, 4-wire	0-25	0-33-1/3	0-22	.75-1.0
60	240/416	3, 4-wire	0-180	0-240	0-158	.75-1.0
60	254/440	3, 4-wire	0-180	0-240	0-158	.75-1.0
400	120	1, 2-wire	0-120	0-160	0-106	.75-1.0
60	120	1, 2-wire	0-25	0-33-1/3	0-22	.75-1.0
60	240	1, 2-wire	0-180	0-240	0-158	.75-1.0
60	254	1, 2-wire	0-180	0-240	0-158	.75-1.0

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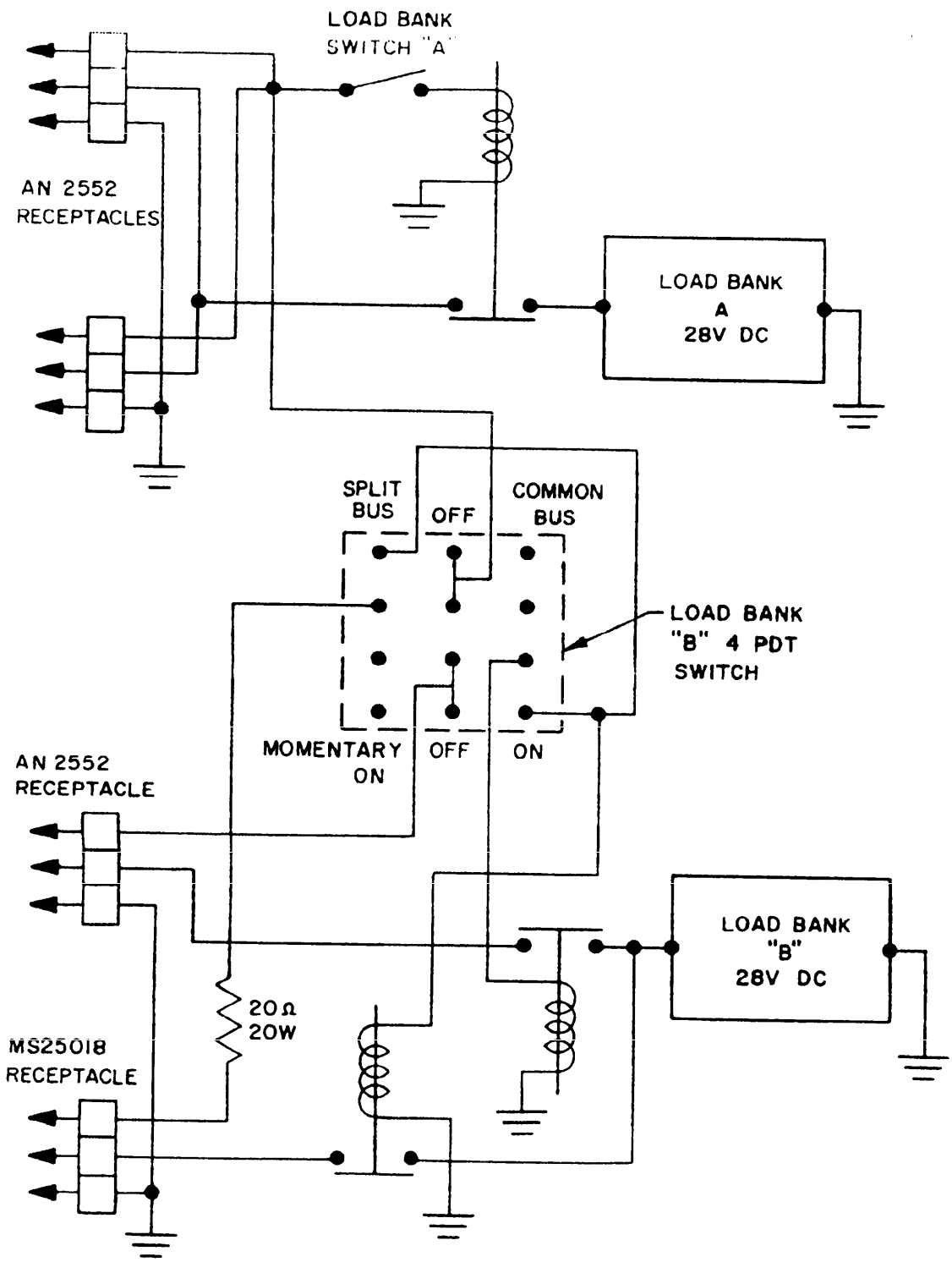


FIGURE I. SCHEMATIC OF COMMON BUS & SPLIT BUS RECEPTACLE ARRANGEMENT

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3.5.8 Illumination. Lighting shall be provided for the meter faces and all controls so that the test set may be operated under any ambient lighting condition. Illumination shall be provided for operation at all voltages and frequencies specified herein. The lighting load shall be supplied by the power source and shall be measured as part of the total load. All lights shall be controlled by a single switch.

3.5.9 Electrical grounding. Electrical grounding shall be accomplished in accordance with MIL-STD-454, Requirement 1. The neutral connector of the 3-phase, 4-wire AC electrical inputs and the negative wire of the 28 volt (V). Direct current (DC) input shall be grounded to the main frame and chassis of the test set by a low resistance electrical connection. An external ground terminal shall be provided so that a separate external ground connection may be added, if desired. This ground terminal shall be connected to the main metal frame and chassis by a low resistance electrical connection.

3.5.10 Dielectric strength. The test set shall be designed and constructed to withstand the voltages specified herein from line-to-ground and between appropriate terminals and other exposed metal parts for the period of time specified herein. Arc resistant materials shall conform to MIL-STD-454, Requirement 26.

a. Circuits of 50V and less: 500V rms for 1 minute or 600V root-mean-square (rms) for 1 second.

b. Circuits of over 50V: twice the rated voltage plus 1,000V rms for 1 minute or 120 percent of the 1-minute value for 1 second.

3.5.11 Mobility. The test set shall be towable over paved highways at 20 miles per hour (mph), graded gravel roads at 10 mph, and type M-6 pierced steel planking conforming to MIL-M-59 at 10 mph. Pneumatic tired wheels shall be used. Pneumatic tires smaller than size 6.00 by 9, six-ply, industrial pneumatic shall not be used. Design shall allow for towing by vehicles having rear-mounted pintle hooks 10 to 30 inches from ground, measured to the center of the opening. The enclosure frame shall serve as a chassis for mounting the running gear and other equipment as necessary to meet the requirements as specified. Lubrication shall be in accordance with MIL-STD-454, Requirement 43.

3.5.11.1 Running gear. The running gear shall be of the 4-wheel, 2-wheel steerable type designed to provide knuckle-type steering with an inside cramping angle of not less than 29 degrees. The front wheel shall be steerable with a hinged drawbar terminating in a standard lunette eye in accordance with MS51336.

3.5.11.2 Pintle. A light-duty, nonquick-release-type pintle assembly in accordance with MS51335 shall be attached at the rear of the equipment for towing.

3.5.11.3 Braking ability. Parking (hand) brakes of the internal expanding or disc type shall be provided.

3.5.11.4 Stability. The test set shall be designed to negotiate side slopes having an incline of 11 degrees without exhibiting a tendency to overturn.

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3.5.12 Forklift capability. The test set shall be designed and constructed for handling and loading by standard forklift trucks.

* 3.5.13 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked, nor otherwise permanently deformed. Fasteners used to retain external panels, doors, and handles shall be of a flush type to minimize damage during use. Fastening devices shall be selected in accordance with MIL-STD-454, Requirement 12. Paragraph 3.5.13 of this specification is not intended to apply to fixed knobs or handles used as a means of opening access doors (i.e., the fire extinguisher compartment door, control panel doors, or covers on top of the load bank). The "flush latch" requirement is intended to eliminate the type of latch, a wing-nut device as used to fasten the side access panels to the A/M24T-8 load bank. Use of wing-nut fasteners is hereby authorized.

3.5.14 Electrical wiring.

3.5.14.1 Wire. Whenever applicable, the wire used throughout the entire test set shall be in accordance with MIL-W-5086, type II.

3.5.14.2 Wiring. Wires shall be so connected to terminals by means of solderless terminal lugs that good connections will be made without soldering. Small components may have soldered terminal connections, provided the soldered joint is not depended on for mechanical strength. All wires or laced bundles of wires shall be securely fastened to the terminal blocks or devices in which they terminate, to prevent the wires from breaking due to vibration. All wires shall be of sufficient size to carry the required current under all environmental conditions without exceeding the temperature rise pertaining to the class of insulation used. (See 3.14.3)

3.5.15 Reflectors. Eight reflectors conforming to MIL-R-12144 shall be provided. There shall be two reflectors at each corner not less than 24 inches nor more than 30 inches above the ground. Reflectors on left and right front corners shall be yellow, reflectors on left and right rear corners shall be red.

3.5.16 Safety devices. Safety considerations shall be as specified in the applicable portions of MIL-STD-454, Requirement 1, paragraphs 4.1, 5, 5.2, 5.10, 5.14, 8, and 8.6. Positive means shall be provided to prevent accidental changing of the input voltage and frequency control during operation. If any part of the test set is susceptible to damage resulting from overvoltage, undervoltage, overfrequency, or underfrequency, means shall be provided to prevent the occurrence of such damage. Fuses, if used, shall conform to MIL-STD-454, Requirement 39, and be the self-indicating type. If fuses are used for protection of any part of the circuit, one spare fuse shall be provided for each fuse used. Meter range switches and load controls shall be interlocked to prevent damage to meters by overloading.

3.5.17 Tiedown rings. Tiedown rings shall be provided on each lower corner of the test set frame to permit the test set to be secured for air transportation and to withstand service acceleration forces.

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3.5.18 Special checking panels and associated cables.

3.5.18.1 Direct-current power distribution panel checker. A special checker conforming to Figure 2 shall be included for testing the MD-3, A-861, MD-3A, and A-897 power distribution panels. Two special cables shall be provided for connecting the checker portion of the test set to the power distribution panel of the generator set. The cables shall consist of the following components.

Cable No. 1	Cable No. 2
Cable 15 foot	Cable 15 foot
Plug MS3106B14S-2P	Plug MS3108B28-11P
Clamp MS2057A6A	Clamp MS2057A16A
Plug MS3101B28-11S	Plug MS3106B28-12S
Clamp MS3057A16A	Clamp MS3057-16A
Bushing MS3420-8	
Bushing MS3420-10	

3.6 Performance.

3.6.1 Operation. The test set shall operate satisfactorily under the following conditions:

a. Operation at temperatures ranging from -54° to +52°C, with +71°C intermittent 20 minute operation.

b. Relative humidity up to 95 percent, including conditions wherein condensation takes place in the form of water and frost.

c. Exposure to salt-sea atmosphere.

d. Fungus growth as encountered in the tropical climates.

e. Rainfall as encountered in any locale.

f. Sand and dust particles as encountered in desert areas.

* 3.6.2 Nonoperation. The test set shall meet the following requirements for storage and airlift:

a. Storage at temperatures from -62° to +71°C without adverse effects to components and subsequent operation.

b. Altitude. Nonoperation from sea level to 50,000 feet above sea level, without hazard resulting from loss in pressure, or adverse effects to components.

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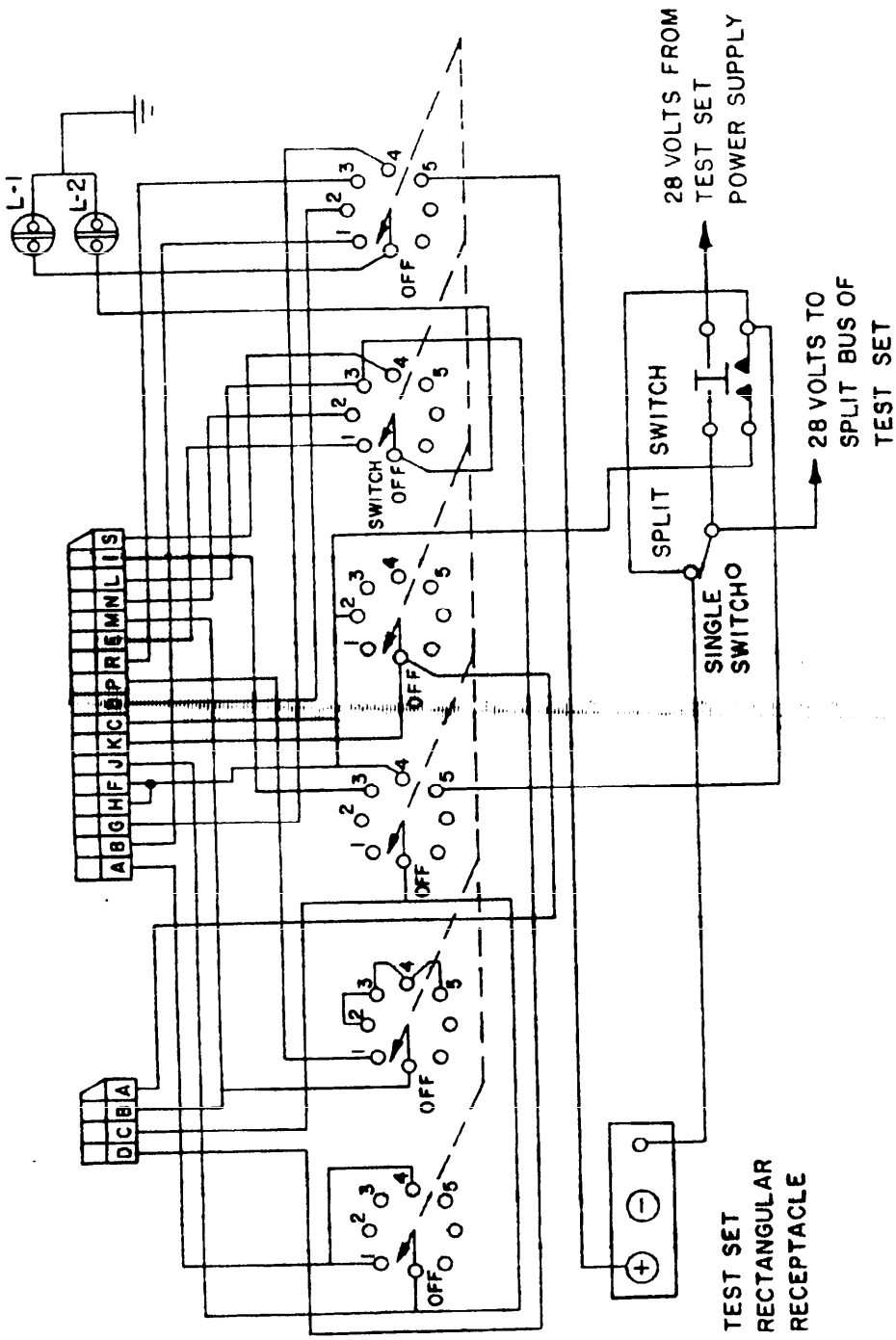


FIGURE 2 A-861 A-897 DC POWER DISTRIBUTION CHECKER

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3.6.3 Acceleration forces.

3.6.3.1 Operable acceleration forces. The assembled test set shall be capable of withstanding a minimum of 3 gravity (g) along each of the three major axes in either direction for a minimum of 0.1 second and shall operate in all respects after such exposure in accordance with all requirements specified herein.

3.6.3.2 Nonoperable acceleration forces. The test set shall be capable of withstanding the following loads to the extent that no component of the test set will become physically separated from its installed position. The test set need not be serviceable after being subjected to such acceleration.

a. A minimum of 8g in either direction applied independently along the logitudinal (horizontal) axis for a minimum duration of 0.1 second.

b. A minimum of 4-1/2g for a minimum duration of 0.1 second in a vertical direction so that the load is imposed on the wheels of the unit.

3.7 Details of components.

3.7.1 Load controls.

3.7.1.1 Impedance and resistance load bank controls. Throughout the specified range, loading shall be controlled by the use of switches, per MIL-STD-454, Requirement 58, relays per MIL-STD-454, Requirement 57, and variable components as necessary to regulate the magnitude of the load within the tolerance specified. It shall be possible to obtain sudden application or removal of any value of selected load by a single load switch after selection is made. Total 3-phase load, load on any one phase, sngle-phase load comprised of the three phases in parallel, and both direct-current loads shall be controllable in this manner. It shall be possible to infinitely vary the load throughout specified range when the switch is closed.

* 3.7.1.2 Switching. All switching and load selection shall be accomplished by switches controlling magnetically operated contactors. Action of relays and contactors shall be positive. All control wiring shall be protected by circuit breakers selected per MIL-STD-454, Requirement 37. Fuses shall not be used.

* 3.7.1.2.1 Control power indicator light. An indicator light shall be installed on the control panel to indicate the presence of a 28V control signal on the 400-cycle, 120/208V input circuits. The lamp circuit shall be connected to pins E and N of the 400-cycle input receptacle. The indicator light shall be selected per MIL-STD-454, Requirement 50.

* 3.7.2 Cooling fans. The cooling fans shall be of the high-speed axial-flow type to maintain minimum dimensions. Fans shall be so wired that they automatically become operative when an applicable load bank is energized. Screened or meshed guards shall be provided over inlets, if applicable, and exhaust air outlets to prevent hazard to operating personnel. All fans shall operate wind switches which shall automatically disconnect the appropriate loads in case of failure of any fan. Cooling fan motors shall be capable of operating satisfactorily for a minimum period of 5,000 hours. Fan motors shall be in accordance with MIL-STD-454, Requirement 46.

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3.7.2.1 Alternating-current impedance bank cooling fans. The impedance load bank shall be cooled by fans powered by identical 28V, DC motors operating from single-phase, 60- and 400-cycle transformer-rectifier units. The transformer-rectifier units shall be capable of driving the motors when operating at all frequencies and voltages specified in 3.5.7. The cooling fan power shall be measured as part of the total load.

3.7.2.2 Resistance bank cooling fans. Cooling fans for the resistance load bank shall operate on 28V DC. The cooling-fan power shall be measured as part of the total load.

* 3.7.3 Enclosure. The enclosure shall be constructed to permit safe operation of the test set during rainy weather, without danger to the operator or the test set. Covers or doors shall be provided for completely protecting the test set against sand or dust when not in use. The covers or doors shall not increase overall dimensions of the test set by more than 25 percent when open, and all corners and edges shall be sufficiently rounded to prevent injury to personnel. All covers or doors and hold-open devices shall be satisfactory for use in the high wind or jet-blast areas by means of rigid bars and chain connections. Top of load bank shall be solid construction, not divided or split in any manner so as to retain rain or moisture, or weaken the structure.

3.7.3.1 Compartment. A compartment shall be provided for storage of the two 15-foot direct-current power distribution panel test cables.

3.7.4 Control console. The control console shall be of welded sheet-steel construction. A collapsible shelf or other work area covered with plastic or other insulating and wear-resistant material, except paint, shall be provided for taking notes and other relevant data. The control console shall be enclosed as specified in 3.7.3.

3.7.4.1 Control panel. The control panel, mounted on the console, shall contain provisions for all controls and instruments used to fulfill the function of the test set. The panel shall be dead front to prevent shock hazard to personnel.

3.7.4.2 Electric meters. The meters shall be of the ruggedized type in accordance with MIL-M-10304, except the initial meter accuracy shall be 1 percent of full-scale value. The meter size shall be not less than 3-1/2 inches in diameter. The color scheme shall be white dial background with black marking and pointer. The power factor meter shall consist of a transducer and a microammeter calibrated to read power factor.

3.7.4.2.1 Power characteristics indication. All meters shall indicate the characteristics of power supplied to the test set by measuring directly and indirectly the applicable voltage, current, power, power factor, and frequency.

* 3.7.4.2.2 Meter accessories. Shunts, calibrating resistors, potential and current transformers, and other transducer units shall be provided where required. Parts shall be selected per MIL-STD-454.

3.7.4.2.3 Scale division multiples and span. Whenever practicable, the value of each scale division shall be either 1, 2, or 5 of the units measured or any decimal multiple or submultiple of these numbers. In the case of multiple-scale meters or of meters for other specialized uses, exceptions to this

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requirement may be necessary, but shall be avoided when reasonably possible. The angle subtended by a scale division shall be not less than 1 degree. Scale span shall be compatible with the stated accuracy requirements and good readability.

3.7.4.2.4 Alternating-current meters. All alternating-current meters shall be suitable for use on 60- and 400-cycles per second (cps), or separate meters shall be provided as required. The range-selector switch of the alternating-current voltmeter shall also govern the voltage coil connections of the wattmeter, power factor meters, and frequency meters. The range-selector switch of the alternating-current ammeter shall also govern the current coil connections of the wattmeter and the power factor meters.

3.7.4.2.5 Frequency meters and wattmeters accuracy. The accuracy of the frequency meters and wattmeter shall not change by more than 1/2 percent of full-scale if the voltage varies from rated value by ±10 percent. The accuracy of the alternating-current voltmeter, ammeter, and wattmeter shall not change by more than 1/2 percent of full-scale if the frequency varies from rated by ±10 percent. During any part of environmental testing, the change in accuracy of any meter shall not exceed 3 percent of full-scale. Following all environmental testing, the permanent change in accuracy of any meter shall not exceed 1 percent.

3.7.4.2.6 Meters detail requirements. Meters shall be in accordance with the following detail requirements:

3.7.4.2.6.1 Alternating-current voltmeter. One alternating-current voltmeter shall be furnished with scales of 150V and 300V..

3.7.4.2.6.2 Alternating-current ammeter. One alternating-current ammeter shall be furnished with scales of 50, 100, 250, and 500 amp.

3.7.4.2.6.3 Frequency meters. One 60-cycle- and one 400-cycle frequency meter shall be provided. The 60-cycle meter shall cover the range of 56 to 64 cycles inclusive and shall be accurate within 0.25 cycle. The 400-cycle meter shall cover the range of 375 to 425 cycles inclusive and shall be accurate within ±1 cycle. A single frequency indicator may be used if the accuracy requirements are met and if the input frequency range selection is accomplished automatically by load selection switching.

3.7.4.2.6.4 Alternating-current wattmeter. One alternating-current wattmeter shall be furnished for indicating power in a single-phase at a time. The meter shall have scales of 5, 10, 25, and 50 killowatts (kw) when operating on 120V, line-to-neutral circuits. On the 240/254V line-to-neutral circuits, the true power shall be double the wattmeter indication.

3.7.4.2.6.5 Power factor (PF) meter. The power factor meter shall consist of a transducer and an indicator. The transducer-indicator shall be calibrated to read power factor at 60 and 400 cycles. The range of the meter shall be from 0.75 lagging to unity power factor. The indicator shall consist of a microammeter in accordance with MIL-M-10304, excent the initial accuracy shall be 1 percent. The power factor meter shall have, at a voltage of 120V ±1 percent, an overall accuracy of ±0.02 PF, through a range of from 0.75 lagging to unity power factor. At a voltage of 120V ±5V, the power factor meter shall have an overall accuracy of ±0.03 PF through a range of from 0.9 lagging to unity power factor, and an overall accuracy of ±0.04 PF through a range of from 0.75 lagging the 0.9 lagging power factor.

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3.7.4.2.6.6 Direct-current voltmeter. Two direct-current voltmeters shall be provided. Each voltmeter shall have a 50V scale.

3.7.4.2.6.7 Direct-current ammeters. Two direct-current ammeters shall be provided. Each ammeter shall have scales of 200, 400, 1000, and 2000 amp.

3.7.4.2.6.8 Phase-sequence indicator. A phase-sequence indicating device incorporating the bright and dark lamp principle shall be provided.

3.7.4.2.6.9 Elapsed time indicator. An electrically driven hourmeter having a range of 0 to 999 hours shall be provided. The indicator shall be of the digit-reading type. The indicator shall show the total time the test set is in use without regard to the magnitude of the load or whether the input current is alternating or direct current.

3.7.4.2.6.10 Selector switch. A selector switch shall be provided so that the alternating current, voltage, power factor, and power of a phase can be measured simultaneously. The switch shall provide the capability of selected readings of any desired phase.

3.7.4.3 Terminal block. An enclosed terminal block shall be provided as part of the control console. It shall contain meter jacks and terminals to facilitate external instrumentation. Terminals shall be of the stud type and shall be constructed of corrosion-resistant material. Studs shall be rigidly mounted in the terminal block. All terminals shall be so marked that the markings will not rub off or become obliterated under normal conditions of use. The terminal block shall not be exposed when the control-panel doors are opened. Circuitry for external instrumentation shall be so designed and constructed that external instruments may be added or removed without de-energizing or damaging the test set.

3.7.5 Electrical hardware and accessory equipment.

3.7.5.1 Input power connections. Power receptacles and terminals shall be mounted in the load bank enclosure. Receptacles and terminals shall be covered with a spring-loaded weatherproof cover. Terminals shall be of corrosion-resistant material of sufficient size to carry rated current of the load. Terminals shall be of the stud type securely fastened to the terminal strip. An insulated knob with threaded insert of the same material and mating thread as the stud shall be provided with each stud so that terminal lugs may be adequately secured without the use of tools. Sufficient clear insulated area shall be provided around all power terminals to permit safe and easy connection of normal lug-terminated cables.

3.7.5.2 Twenty-eight-volt, direct-current input. Power shall be supplied to the 28V DC resistive load banks through three external power receptacles in accordance with AN2552, and one external power receptacle in accordance with MS25018 for use on split-bus operation. In addition, two terminal strips shall be provided for the 28V DC resistive loads so that loads may be connected through lug-terminated cables.

3.7.5.3 Sixty- and four-hundred-cycles-per-second, alternating current input. Power shall be supplied to the alternating-current impedance load bank through two external power receptacles in accordance with AN3114, and one male receptacle with pin arrangements and dimensions identical to the male plug show on Drawing 55C6182. In addition, two terminal strips shall be provided for the alternating-current impedance loads so that loads may be connected

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3.7.5.4 Extension light. One 28-V extension light with 60 feet of cable shall be provided. Space shall be provided for permanent storage of the extension light in the test set. At least two output receptacles to supply power to the extension light shall be provided on the load bank but not on the control panel. When the light is connected to the load bank and energized, the power drawn shall be measured as part of the total load.

3.7.5.5 Fire extinguisher. A fire extinguisher conforming to MIL-E-5220 shall be mounted on the test set in a location readily accessible from the control console.

3.7.6 Handbook and record compartment. A compartment having minimum dimensions of 8 X 11-1/2 by 2 inches shall be provided inside the test set enclosure for the purpose of containing equipment operation handbooks and equipment maintenance and inspection records.

* 3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-1000 shall govern the manufacturer's part numbers and changes thereto. (See also 3.4.3)

* 3.9 Screw threads. Screw threads shall conform to Handbook H28 and MIL-STD-454, Requirement 12.

* 3.10 Electromagnetic interference. The test set shall meet the requirements specified in MIL-STD-461/462.

3.11 Dimensions. The test set shall not occupy a volume greater than 225 cubic feet, assuming a box or crate having this inside volume could completely enclose the test set. Overall height shall not exceed 5 feet. These dimensions apply to the test set with door closed.

3.11.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the test set, they shall be held or limited accordingly.

3.12 Weight. The total weight of the test set shall not exceed 3,700 pounds.

3.13 Finishes and protective coatings.

* 3.13.1 Cleaning, painting, plating, anodic films, and chemical treatment. Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with MIL-STD-808. The test set shall be finished in yellow, film designation DG. Parts subject to high temperatures shall be finished in accordance with finish code FF-903.

3.14 Operational markings.

3.14.1 Instructions. Brief instructions for operation, together with a complete electrical-system wiring diagram, shall be permanently located on the control panel specified herein. The instructions shall be concise and adequate to permit all degrees of loading as specified herein without damage to the test set. The instructions shall include the type of lubricant recommended for the cooling fan motors throughout temperatures ranging from -54° to +71°

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Centigrade (C). The instructions and diagrams shall be such that oil, dirt, exposure to light, et cetera, will not affect their legibility or cause them to be erased. The wiring diagram shall be such that any wire can be readily found by reference to the diagram. Silk screening, if used, shall be accomplished in like manner as on the control panel marking. Plastic laminated sheets loosely attached by rivets, not permanently affixed to the panel door, shall not be acceptable.

3.14.2 Conversion instructions. Instructions for accomplishing the conversion and method of loading specified in 3.5.4.1 shall be provided and shall be located on the control panel. A caution plate shall be provided to warn the operator to determine that the load bank is connected to accept the desired power source before applying power.

3.14.3 Wire identification. Each wire shall be identified at not more than 15-inch intervals throughout its entire length. The wires shall also be identified within 3 inches of the terminating joint or junction. The identification shall be permanently imprinted on the wire or sleeve and shall be so located that it can be easily read without removal of ties, clamps, or coverings. This marking shall be in accordance with MIL-W-5088. Required identification includes wire number; wire segment letter; wire size number; and ground, phase, or the thermocouple letters. Use of unit number, circuit function letter, armament special system number, and circuit designation letter is optional.

* 3.14.4 Indicator light. The indicator light specified in 3.7.1.2.1 shall be identified as to its function by a circular metal plate or metallic decal on the control panel. Silk screening, if used, shall be subject to approval by the procuring activity.

3.15 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.15.1 Request for nomenclature. Request for nomenclature shall be as specified in the contract or purchase order.

* 3.16 Workmanship. Workmanship shall conform to MIL-STD-454, Requirement 9.

3.16.1 Riveting. Riveting operations shall be performed to insure tightness and satisfactory heading of rivets.

3.16.2 Cleaning. The test set shall be thoroughly cleaned and loose, spattered, or excess older; metal chips; and other foreign material shall be removed during and after final assembly.

* 3.16.3 Castings and forgings. All castings shall be of uniform quality, free from blowholes, porosity, hard spots, shrinkage defects, cracks or other injurious defects. In no event shall processes such as peening or plugging be used on castings or forgings for reclaiming any parts of the test set. Castings shall be as specified in MIL-STD-454 Requirement 21.

* 3.17 Color. The load bank outside surface color shall be flight line yellow per FED-STD-595.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the test set shall be classified as follows:

- a. Preproduction testing See 4.4

b. Acceptance tests See 4.5

4.3 Test conditions.

4.3.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of approximately 29.92 inches mercury (Hg) and at a temperature of approximately 25°C. Where tests are made with atmospheric pressure or temperature substantially different from these values, proper allowance shall be made for the change in instrument reading.

4.3.2 Apparatus. Insofar as practicable, apparatus used in conjunction with the testing specified herein shall be of the laboratory-precision type and shall be calibrated at intervals properly spaced to assure laboratory accuracy. All apparatus used shall be acceptable to the procuring activity.

4.4 Preproduction testing.

4.4.1 Preproduction test sample. Two test sets shall be subjected to the tests specified in 4.4.3, conducted in accordance with Table I in the order specified.

4.4.1.1 Test plan. Subsequent to award of a contract, the contractor shall submit a test plan for approval by the procuring activity. The test plan shall consider each of the specified performance and testing requirements and shall represent the contractors understanding of these requirements.

4.4.2 Preproduction test report. After the preproduction tests are completed, three completed copies of a test report in accordance with MIL-STD-831 shall be supplied to the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be submitted as an attachment accompanying the preproduction test report (same number and sort of copies) or shall be included as part of that report.

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time, miles, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded.

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TABLE I

PREPRODUCTION TEST SAMPLES

Test Sequence	Samples		Applicable Paragraph
	No 1	No 2	
Examination of product	X	X	4.6.1
Serviceability	X	X	4.6.3
Electromagnetic interference	X		4.6.6
Dielectric strength	X		4.6.5
Acceleration loadings, operable	X		4.6.4.1
Initial calibration	X		4.6.2
Environmental	X		4.6.8
Linearity of reactive loading elements		X	4.6.4.6
Cooling fans		X	4.6.7
Mobility		X	4.6.9
Final calibration	X	X	4.6.10
Dielectric strength		X	4.6.5
Acceleration loadings, nonoperable		X	4.6.4.2
Phase conversion		X	4.6.11
Reliability	X	X	4.6.12
Maintainability	X	X	4.6.13

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b. Description of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Description of the engineering reasoning behind any corrections made, to be made on production items, or proposed to be made and behind the predicted effectiveness of those corrections.

d. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item.

f. Estimates (±5 percent of actual experienced) of manhours required for each maintenance and servicing action during the tests. A brief description of the qualifications and experience of the personnel involved shall be included and shall be adequate to permit comparison to the personnel anticipated in similar fieldwork.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6.

4.5 Acceptance tests. Acceptance tests shall consist of the following:

a. Individual tests	See 4.5.1
b. Periodic sampling plan and tests	See 4.5.2

4.5.1 Individual tests. Each test set shall be subjected to the following tests as described under 4.6:

a. Examination of product	See 4.6.1
b. Initial calibration	See 4.6.2

* 4.5.2 Periodic sampling plan and tests. The sampling plan shall be in accordance with MIL-STD-105 for general inspection level II. The acceptance quality level shall be four (4) percent for all the tests specified in Table II. Major and minor defects shall be as specified in MIL-STD-105.

4.5.2.1 Rejection and retest. When one item selected from a production run fails to meet the specification, items still on hand or later produced shall not be accepted until the extent and cause of failure are determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.2 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure, but final acceptance of the items on hand or produced later shall not be made until it is determined that they meet all the requirements of the specification.

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TABLE II - PERIODIC SAMPLING TESTS

Examination or Test	Requirement Paragraph	Method Paragraph
a. Acceleration loading operable	3.6.3.1	4.6.4.1
b. Alternating current loads	3.5.7	4.6.4.3
c. Direct current loads (resistance	3.5.5 & 3.5.7	4.6.4.4
d. Split-bus loading	3.5.5.1	4.6.4.5
e. Dielectric strength	3.5.10	4.6.5

4.5.2.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods.

4.6.1 Examination of product. The test set shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship, marking, dimensions, and accessibility of components.

4.6.2 Initial calibration. The test set shall be subjected to the tests specified in 4.6.4.3, 4.6.4.4, and 4.6.4.5, and 4.6.4.7 for initial calibration.

4.6.3 Serviceability. The test set shall be inspected and evaluated to determine ease of maintenance, servicing, and operation.

4.6.4 Loading.

4.6.4.1 Acceleration loadings, operable. The ability of the test set to withstand the acceleration forces specified in 3.6.3.1 shall be demonstrated as follows:

a. The test set shall be securely fastened to a suitable pad by its tiedown rings.

b. The g-loads specified in 3.6.3.1 shall be applied twice along the axes specified.

c. An oscillograph recording of g-force versus time shall be recorded for each impact.

d. The test set shall then be inspected for compliance with 3.6.3.1.

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4.6.4.2 Acceleration loadings, nonoperable. The ability of the test set to withstand the acceleration forces specified in 3.6.3.2 shall be demonstrated as follows:

a. The test set shall be securely fastened to a suitable pad by its tiedown rings.

b. The g-loads specified in 3.6.3.2 shall be applied twice along the axes specified.

c. An oscillograph recording of g-forces versus time shall be recorded for each impact.

d. The test set shall then be inspected for compliance with 3.6.3.2.

4.6.4.2.1 Acceleration test waiver. The test for the requirement specified in 4.6.4.2 may be waived upon submission of engineering data demonstrating ability to withstand these loadings. Whenever this test is conducted, expensive or sensitive meters that would be damaged may be removed from the test set. However, such instruments must be replaced during the test by a dummy instrument of equivalent size, weight, and provision for mounting.

4.6.4.3 Alternating-current loads (resistance and inductive reactance). Each phase of the alternating-current loads shall be connected to a suitable power source, and loads shall be applied from zero to full-load at the required power factors, voltages, and frequencies specified in 3.5.7. Sufficient readings shall be taken to demonstrate that the test set will provide the loading at the frequencies and voltages specified in 3.5.7 throughout the power factor range in increments of 0.05 PF. Volts, amperes, kilowatts, power factor, and frequency shall be recorded for each step of the test. Simultaneous readings shall be taken from the test set meters and laboratory standard meters of ±1/4 percent accuracy or better.

4.6.4.4 Direct-current loads (resistance only). The direct-current loads shall be connected to a source of power as specified in 3.5.7. It shall be demonstrated that each section (see 3.5.5) can absorb loads proportional to incremental current changes of 1 amp, from 0 to 1,500 amp. Volts and amperes shall be recorded for each step.

4.6.4.5 Split-bus loading. The direct-current loads shall be connected to a suitable power supply and split-bus operation shall be demonstrated. The direct-current loads shall be connected for common-bus operation and a load applied to the source generators. Voltmeter and ammeter readings shall be recorded. The jet-starting signal of the source shall be applied and test set meter readings shall again be recorded. The two separate loads shall then be varied and meter readings shall be recorded to thoroughly demonstrate the regulated voltage and the constant current characteristics of the source generators.

6.4.6. Linearity of the reactive loading elements. The linearity of the ctive elements shall be demonstrated by providing contractor-prepared test . The data shall include saturation curves of load current plotted against density. The currents to which the reactive elements will be subjected be furnished and their respective positions on the saturation curve e indicated.

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4.6.4.7 Special checking panels and associated cables. A thorough continuity check shall be made on the direct-current power distribution panel checker to insure that the checkers and associated cables are in accordance with Figure 2.

4.6.5 Dielectric strength. The test set shall be subjected to voltages as specified in 3.5.10 at commercial frequency from line-to-ground and between appropriate terminals and other exposed metal parts for the period of time specified. This test shall be conducted at normal room temperature and standard atmospheric conditions. There shall be no insulation breakdown.

4.6.6 Electromagnetic interference. The test set shall be tested for radiated interference in accordance with MIL-STD-461/462 for Class IIIC equipment. The antenna shall be placed 1 meter from outer perimeter of the unit at the point of maximum interference.

4.6.7 Cooling fans. It shall be demonstrated that when ventilation from a cooling fan fails or ventilating doors are closed, the load shall be automatically disconnected from the line by means of the wind switch.

4.6.7.1 Cooling fan motor life test. Ten cooling fans of the type to be furnished on the test set shall each be run for 2,500 hours prior to delivery of the first article. The motors shall be operated in an ambient temperature equal to that experienced during conditions of full load on the test set with the outside ambient air at 24° ±1°C. The following measurements shall be made prior to the beginning of the test run:

a. Commutator diameter and concentricity.

b. Brush length.

c. Diameter and concentricity of motor shaft at the point of contact with the bearings.

d. Diameter and concentricity of bearings.

4.6.7.1.1 Life test - extrapolation. At the time of submission of the first article and first article test report, the motor shall have run 2,500 hours. The first article test report shall include the specified measurements made at the beginning of the test and also those made at the time of submission of the test report; the running time shall be at least 2,500 hours. Extrapolation of anticipated wear of 5,000 hours shall be made based on the readings obtained and on the allowable amount of wear permitted for a useful life. An acceptable extrapolated life of 5,000 hours shall be one of the conditions for approval of the first article test set.

4.6.7.2 Test data. If test data is available on the fans that can substantiate a life of 5.000 hours, such data may be submitted to the procuring activity for approval immediately following the award of any contract. These data may be submitted for consideration of waiver of this test. If the data submitted is not satisfactory, the tests specified in 4.6.7.1 shall be run by the contractor.

4.6.8 Environmental. The test set shall be subjected to the climatic conditions of MIL-E-4158 in accordance with MIL-STD-810 for the following tests.

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4.6.8.1 High temperature. The high temperature test shall be conducted in accordance with Method 501, Procedure I. The test set shall be operated at +52°C for at least 30 minutes to verify satisfactory performance of the loading requirements of 3.5.7.

4.6.8.2 Low temperature. The low temperature test shall be conducted in accordance with Method 502, Procedure I.

4.6.8.3 Humidity. The humidity test shall be conducted in accordance with Method 507, Procedure I.

4.6.8.4 Salt-fog. The salt-fog test shall be conducted in accordance with Method 509, Procedure I for 50 hours.

4.6.8.5 Fungus. The fungus test shall be conducted in accordance with Method 508, Procedure I. If all materials used in the test set are either treated with a fungus-resistant material, or are hermetically sealed, the fungus-resistance test may be waived by the contracting officer. Certification of fungus resistant materials must be submitted by the contractor, to the procuring activity not later than 30 days prior to preproduction testing. Operation of the test set at the end of the test is required.

4.6.8.6 Rain. The rain test shall be conducted in accordance with Method 506, Procedure I. The test set shall be operated during the last 10 minutes of the 30 minutes rain simulation.

4.6.8.7 Sand and dust. The sand and dust test shall be conducted in accordance with Method 510, Procedure I.

4.6.8.8 Altitude. The altitude test shall be conducted in accordance with method 500, Procedure I. The test set shall be operated at the 10,000 foot altitude simulation for at least 30 minutes.

4.6.8.9 Explosive atmosphere. The explosive atmosphere test shall be conducted in accordance with Method 511, Procedure I.

4.6.9 Mobility. The test set shall be towed by a truck tractor or a vehicle having a pintle height of 10 to 30 inches, as applicable, over level paved highways at an average speed of 20 mph, maximum speed 25 mph, for a distance of 100 miles continuously. The test set shall then be towed over graded, gravel roads at an average speed of 10 mph, maximum speed 12.5 mph, for a distance of 50 miles. The test set shall then be towed over a course consisting of obstructions 1 inch in height, regularly spaced at 6-foot intervals and at right angles to the direction of travel of the vehicle at an average speed of 10 mph for a distance of 5 miles. At least 50 right circle turns, 50 left circle turns both at the maximum cramping angle, and 25 sudden stops from the speeds specified shall be accomplished during the test. At the conclusion of the test, the running gear shall be examined for excessive wear or other damage. Excessive wear or other damage to the running gear shall be cause for rejection. The force required to tow the test set shall not exceed 50 lb per ton of weight. Additional mobility tests shall be conducted as specified in 4.6.9.1 and 4.6.9.2.

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4.6.9.1 Braking ability. The parking brakes shall lock the wheels so that the wheels will skid and not roll on dry, level concrete when the equipment is subjected to a towing force sufficient to move the equipment. This shall be demonstrated 15 consecutive times. Brake wheels turning during this test shall be cause for rejection.

4.6.9.2 Stability. The test set shall be towed in both directions along side slopes having an incline of 11 degrees without exhibiting any tendency to overturn.

4.6.10 Final calibration. For final calibration, repeat the tests specified in 4.6.2.

4.6.11 Phase conversion. It shall be demonstrated that the output phase conversion as specified in 3.5.4.1 can be effected.

4.6.12 Reliability testing. Reliability testing shall be performed to verify that the load bank meets the requirements of 3.5.1. The load bank shall be operated at not less than 1/3 of the maximum load capability requirement 3.5.7. The accept-reject criteria and tests shall be in accordance with MIL-STD-781, Test Plan IV, Test Level A-1. A failure is defined as any need to recalibrate by any adjustment not considered an operating control, or any replacement of component parts afer the test is initiated.

4.6.13 Maintainability demonstration. Compliance with the maintainability requirement of 3.5.2 shall be demonstrated by a maintenance procedure, maintenance task selection and task performance in accordance with MIL-STD-471, appendixes A and B, Test Method 1-A. The maintenance demonstration shall be performed to the lowest discrete component level. (i.e., relays, transformers, reactors, meters.)

4.6.14 Heat run. The test set shall be connected to a suitable power source and operated at full load of each leading requirement listed in 3.5.7 for equal time duration for a total time of not less than eight hours. During these runs all temperatures in the load bank shall remain at a safe operating level, and the temperature of the enclosure, controls, and control panel shall not exceed those specified in MIL-STD-454, Requirement 1. Paragraph 8.6 applies, except to the exhaust air and meshed/screened exhaust air outlets.

4.7 Inspection of the preservation, packaging, and packing. The inspection of the preservation, packaging, and packing shall be in accordance with the instructions in Section 5. or the specifications referenced therein.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A. Each test set shall be preserved in accordance with Method III of MIL-P-116. Before closing and latching cabinet doors and replacing cabinet panels, instrument glass shall be padded with suitable cushioning material as protection against shock and vibration. Tape conforming to PPP-T-60 shall be applied to all slams, doors, panels, and other openings.

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5.1.2 Level C. Test sets shall be preserved and packaged against corrosion, deterioration, and physical damage during shipment from supply source to first domestic receiving activity for immediate use. The level conforms to application carrier rules and regulations and may be the suppliers' commercial practice.

5.2 Packing.

5.2.1 Level A. Test sets preserved in accordance with 5.1.1 shall be packed in bolted crates conforming to MIL-C-3774. All crates shall be provided with rubbing strips with a nominal depth of 2 inches. Rubbing strips shall be attached to the container in a manner that will provide for side forklift entry. This requires two openings, devoid of rubbing strips, each a minimum of 24 inches wide and spaced approximately 30 inches on center. Rubbing strips shall be set back not less than 4 inches from the lower ends of the skids. When these requirements for rubbing strips disagree with the requirements for rubbing strips specified in the applicable crate specifications, the requirements of the specification shall prevail.

5.2.2 Level B and air. Test sets preserved in accordance with 5.1.1 do not require crating or external containers for shipment.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), the unit and shipping container shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

TEST SET, ELECTRICAL POWER A/M24T-8( )

5.4 Packaging performance tests. Packaging performance tests shall be in accordance with Table III of MIL-P-116.

6. NOTES

6.1 Intended use. The A/M24T-8( ) test set is intended for use in testing all power supplies that are used for electrical ground support.

6.2 Ordering data. Procuring documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of preservation and packaging, and packing (see 5.1 and 5.2).

c. Special shipment marking (see 5.3).

d. Test plan (see 4.4.1.1).

6.3 Mean-time-between-failures. The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failures.

6.4 Nomenclature. The parentheses appearing in the nomenclature will be dropped and replaced by a number or letter, to indicate the particular design. Nomenclature for the individual components where required will be furnished upon application to the procuring activity (see 3.15.1).

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6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

[ ] YES [ ] NO *(If "Yes", in what way?)*

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